MN TECH ACADEMY

MongoDB Compass

Complete Beginner Step-by-Step Practical Guide

Course: Full Stack Web Development

Topic: MongoDB + MongoDB Compass

Level: Beginner

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1 MongoDB Kya Hai?

MongoDB ek **NoSQL database** hai jahan hum apni website/application ka data store kar sakte hain.

Example: Agar hamare paas Student Registration Form hai:

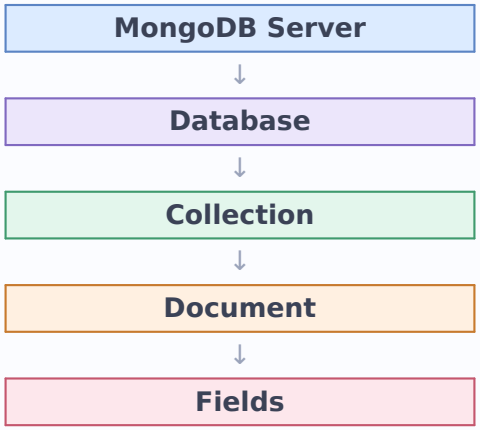
- Name
- Email
- Phone
- Course

to hum ye data MongoDB mein save kar sakte hain. MongoDB mein data generally **documents** ki form mein save hota hai.

```
{
  "name": "Ali",
  "email": "ali@gmail.com",
  "course": "Web Development"
}
```

2 MongoDB Mein Basic Structure

MongoDB ko samajhne ke liye ye structure yaad rakhein:



Example: MongoDB → `studentDB` → `students` → Student document → name, email, phone, course

3 Practical Start Karne Se Pehle Kya Install Hona Chahiye?

Student ke laptop mein ye cheezen honi chahiye:

Required Software

1. MongoDB Community Server
2. MongoDB Compass
3. Node.js
4. VS Code

Check Node.js

Windows mein:

1. Start Menu open karein.
2. Search karein: `cmd`
3. Command Prompt open karein.
4. Type karein: `node -v`
5. Enter press karein.

Agar version aa jaye, for example:

```
v24.x.x
```

to Node.js installed hai. Phir type karein:

```
npm -v
```

Agar version aa jaye to npm bhi installed hai.

4 MongoDB Server Installed Hai Ya Nahi Check Karein

Windows Start Menu open karein aur search karein: `Services`

Services application open karein. Ab list mein search karein: `MongoDB Server` ya `MongoDB`

Agar MongoDB service available hai to uska status check karein.

Agar Status "Running" Hai

MongoDB server already running hai. Aap next step par ja sakte hain.

Agar Status "Stopped" Hai

MongoDB par **Right Click → Start** click karein.

5 Agar MongoDB Start Karte Waqt Error 1053 Aaye

Agar Start par click karne ke baad `Error 1053` aaye to baar baar Start button click na karein.

Iska matlab MongoDB Windows service properly start nahi ho rahi. Pehle MongoDB installation/service ko troubleshoot karein.

> **Important** Agar MongoDB service start nahi hogi to Compass mein `ECONNREFUSED 127.0.0.1:27017` jaisa error aa sakta hai. Isliye pehle MongoDB Server ko properly running karna zaroori hai.

6 MongoDB Compass Open Karein

Windows Start Menu open karein aur search karein: `MongoDB Compass`

Compass open karein. Aapko **New Connection** screen nazar aa sakti hai.

7 MongoDB Local Server Se Connect Karna

Connection string / URI wale box mein ye likhein:

```
mongodb://localhost:27017
```

Phir **Connect** par click karein.

8 Successful Connection Kaise Pata Chalega?

Agar connection successful ho gaya to Compass mein MongoDB ka interface open ho jayega. Aapko databases ka area nazar aayega.

Ab hum apna pehla database banayenge.

9 Apna Pehla Database Create Karein

MongoDB Compass mein **Create Database** par click karein. Do fields aa sakti hain:

Database Name

```
studentDB
```

Collection Name

```
students
```

Phir **Create Database** par click karein.

10 Database Aur Collection Samjhein

Humne banaya: `studentDB` — ye hamara **database** hai. Iske andar `students` **collection** hai.

Students collection ke andar hum students ka data save karenge.

11 Collection Open Karein

Left side par `studentDB` par click karein. Phir `students` collection par click karein.

Ab collection ka page open hoga.

12 Pehla Student Add Karein

Collection ke andar **Add Data** par click karein. Phir **Insert Document** select karein.

Aapko ek document editor nazar aayega. MongoDB usually `_id` automatically generate karta hai. Hum apne fields add karenge.

```
{
  "name": "Ali",
  "email": "ali@gmail.com",
  "phone": "03001234567",
  "course": "Web Development"
}
```

Phir **Insert** par click karein.

13 Data Successfully Save Hua Ya Nahi?

Insert ke baad collection mein document nazar aayega. Aapko kuch is tarah data nazar aa sakta hai:

```
name: Ali
email: ali@gmail.com
phone: 03001234567
course: Web Development
```

MongoDB ne automatically `_id` bhi create ki hogi.

14 Dusra Student Add Karein

Dobara **Add Data → Insert Document** par click karein. Ab ye insert karein:

```
{
  "name": "Sara",
  "email": "sara@gmail.com",
  "phone": "03111234567",
  "course": "Frontend Development"
}
```

Ab collection mein 2 students honge.

15 Teesra Student Add Karein

Practice ke liye:

```
{
  "name": "Ahmed",
  "email": "ahmed@gmail.com",
  "phone": "03221234567",
  "course": "Full Stack Development"
}
```

Insert karein. Ab `studentDB` → `students` mein **3 documents** hone chahiye.

16 MongoDB Document Mein Field Kya Hoti Hai?

```
{
  "name": "Ali",
  "email": "ali@gmail.com"
}
```

Yahan `name` aur `email` **fields** hain. Aur `Ali` aur `ali@gmail.com` un fields ki **values** hain.

17 Existing Data Edit Karna

Kisi document ko edit karne ke liye:

1. Student document locate karein.
2. Document ke paas Edit option / pencil icon locate karein.
3. Us par click karein.
4. Required value change karein.

Example: `course: Web Development` ko change karke `course: Full Stack Development` kar dein. Phir changes save karein.

18 Document Delete Karna

Agar kisi document ko delete karna ho:

1. Required document locate karein.
2. Delete/trash option par click karein.
3. Confirmation aaye to carefully confirm karein.

> **Important** Delete ki hui information ko bina backup ke recover karna easy nahi hota. Isliye practice ke waqt sirf test data delete karein.

19 MongoDB Compass Mein Search Karna

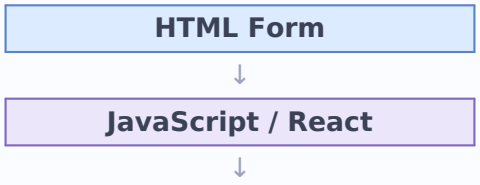
Agar collection mein bohat saare students hon to filter use kar sakte hain. Example:

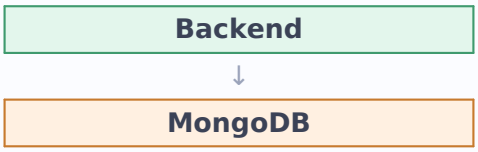
```
{ "course": "Web Development" }
```

Iska matlab: sirf wo documents show karein jinka course Web Development hai.

20 Ab Real Web Project Ki Tayyari

Ab hum MongoDB Compass manually use karna samajh chuke hain. Lekin real website mein user khud MongoDB Compass open nahi karta. Website ka in tareeqe se connection hota hai:





Isliye next practical mein hum Node.js aur Express ke through MongoDB connect karenge.

21 Backend Project Create Karein

VS Code open karein. Terminal open karein: **Terminal → New Terminal**

Ek folder mein project create karein. Example:

```
mkdir mongo-student-project
```

Phir:

```
cd mongo-student-project
```

22 Node Project Initialize Karein

Terminal mein run karein:

```
npm init -y
```

Isse `package.json` file create hogi.

23 Required Packages Install Karein

Terminal mein run karein:

```
npm install express mongoose cors
```

Development ke liye:

```
npm install --save-dev nodemon
```

24 Backend File Create Karein

VS Code mein `server.js` file create karein. Is file mein backend code likha jayega.

25 MongoDB Connection

Local MongoDB ke liye connection string:

```
mongodb://localhost:27017/studentDB
```

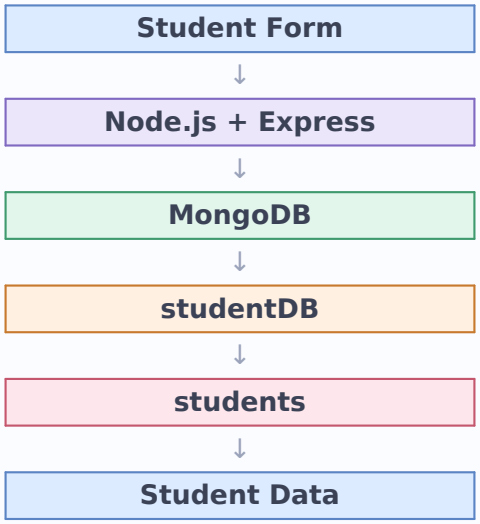
- `localhost` ka matlab hamara apna computer hai.

- `27017` MongoDB ka default port hai.
- `studentDB` hamara database hai.

26 Important Concept

Compass MongoDB database ko manage/view karne ke liye GUI tool hai. **MongoDB Server** actual database service provide karta hai. **Node.js/Express** application MongoDB se connect karke website ka data save/retrieve karti hai.

Simple flow:



27 Common Error: ECONNREFUSED

Error Agar Compass mein ye error aaye: `ECONNREFUSED 127.0.0.1:27017`

To sabse pehle check karein:

1. Windows Services open karein.
2. MongoDB service locate karein.
3. Check karein ke service `Running` hai ya nahi.
4. Agar stopped hai to start karein.

Agar start karte waqt Error 1053 aaye to MongoDB service/installation ko troubleshoot karein.

28 Common Mistakes

1. MongoDB Server installed nahi hai.
2. MongoDB Server installed hai lekin service running nahi.
3. Compass installed hai lekin MongoDB Server nahi.
4. Wrong connection URI enter karna. Local MongoDB ke liye normally `mongodb://localhost:27017` use kiya jata hai.
5. Port 27017 kisi aur configuration mein changed ho.
6. Students MongoDB Compass ko hi MongoDB Server samajh lete hain.

> **Tip** Yaad rakhein: **Compass = GUI**, **MongoDB Server = Database service**.

29 Student Practice Task

Task: Student Registration Database

MongoDB Compass mein ye create karein:

- Database: `academyDB`
- Collection: `students`

Kam az kam 5 students ka data insert karein. Har document mein ye fields hon: `name`, `email`, `phone`, `course`, `age`

```
{
  "name": "Ayesha",
  "email": "ayesha@gmail.com",
  "phone": "03001234567",
  "course": "Web Development",
  "age": 20
}
```

30 Practice Questions

Students khud perform karein:

Q1	Naya database create karein: `academyDB`
Q2	Collection create karein: `students`
Q3	5 students insert karein.
Q4	Ek student ka course edit karein.
Q5	Ek student delete karein.
Q6	Sirf `Web Development` course wale students ko filter karein.
Q7	Ek new field add karein: `city`, for example `"city": "Karachi"`

31 Final Checklist

Practical complete karne se pehle student ye checklist check kare:

- ☐ MongoDB Community Server installed
- ☐ MongoDB Compass installed
- ☐ Node.js installed
- ☐ MongoDB service running

- [ ] Compass successfully connected
- [ ] `studentDB` database created
- [ ] `students` collection created
- [ ] At least 3 documents inserted
- [ ] Document edit kiya
- [ ] Document delete kiya
- [ ] Filter use kiya
- [ ] MongoDB ka Database → Collection → Document concept samajh aaya

32 Most Important Things To Remember

MongoDB Connection

```
mongodb://localhost:27017
```

Database Example

```
studentDB
```

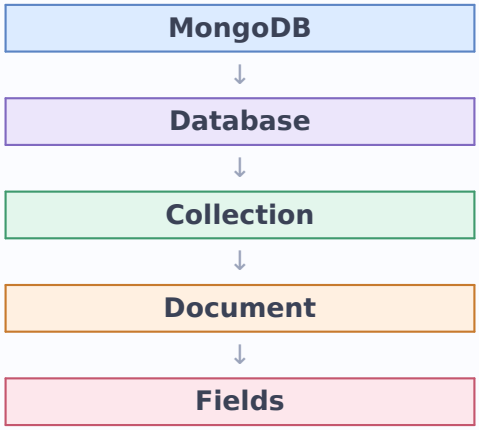
Collection Example

```
students
```

MongoDB Document Example

```
{
  "name": "Ali",
  "email": "ali@gmail.com",
  "course": "Web Development"
}
```

Basic Structure



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Practical Goal

By the end of this lesson, students should be able to:

Install → Connect → Create Database → Create Collection → Insert Data → Read Data → Edit Data → Delete Data → Filter Data